

Project Report

GLOBAL SUPPLY CHAIN ANALYTICS

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Tools Used: MS Excel, Tableau

Course: Data Science & Analytics

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Project Overview:

This project analyses supply chain performance using Tableau dashboards. It focuses on shipment efficiency, supplier performance, inventory management, and customer behaviour to identify operational risks and improvement opportunities.

Data Model Diagram:

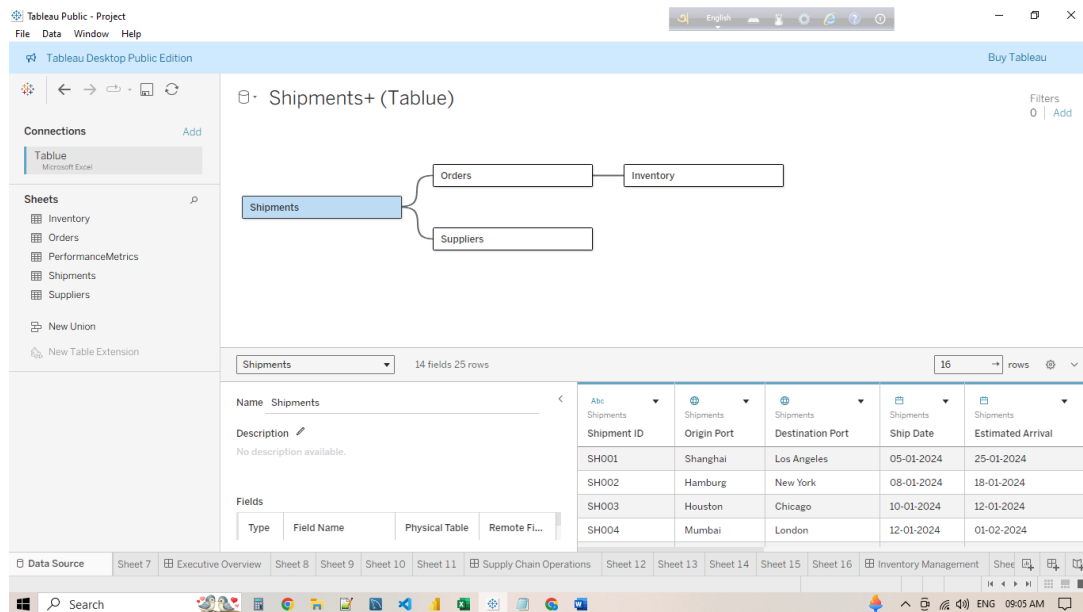


Figure 1: Logical Data Model Structure

The data model integrates Shipments as the primary fact table connected to Orders, Suppliers, and Inventory tables. Relationships ensure accurate aggregation and enable multi-domain analytics across supply chain operations.

Dataset Description:

Table	Description
Shipment	Contains shipment details
Orders	Customer orders data
Inventory	Warehouse stock info
Suppliers	Supplier performance

Calculations Used:

Delivery Delay (Days) = DATEDIFF ('day', [Estimated Arrival], [Actual Arrival])

On-Time Delivery Flag = IF [Actual Arrival] <= [Estimated Arrival]
THEN "On Time"
ELSE "Delayed"
END

On-Time Delivery Rate = SUM (IIF ([Actual Arrival] <= [Estimated Arrival],1,0)) /
COUNT ([Shipment ID])

Total Shipment Value = SUM ([Shipment Value])

Average Shipment Value = AVG ([Shipment Value])

Monthly Shipment Value = {INCLUDE DATETRUNC ('month'[Estimated Arrival])
SUM ([Shipment Value])}
Stock Change = SUM ([Current Stock]) – LOOKUP (SUM ([Current Stock]), -1)
Supplier Average On-Time Rate (LOD) = {FIXED [Supplier Name]:
AVG ([On Time Delivery Rate])}
Customer Total Value = SUM ([Order Value])
Customer Priority = IF SUM ([Order Value]) > 50000 THEN "High"
ELSEIF SUM ([Order Value]) > 20000 THEN "Medium"
ELSE "Low"
END

Dashboard Navigation Guide:

1. Executive Overview
 - Displays KPIs
 - Shows Trend Line Chart
 - Average Delay Days
2. Supply Chain Operations
 - Supply Route
 - Delay Analysis
3. Inventory Management
 - Stock Comparison
 - Stock Level
 - Capacity Comparison
 - Record Alert

Key Insights:

- Only **68%** shipments delivered on time.
- Delay days average **1.75**
- High-value shipments often delayed.
- Few customers generate majority of revenue.
- Supplier performance varies widely.
- Some suppliers consistently delay shipments.
- Inventory in Warehouses is not balanced

Recommendations:

- Improve logistics tracking for delayed routes
- Balance inventory across warehouses
- Prioritize high-value customers

Conclusion:

This project demonstrates how data visualization can improve supply chain decision-making. Using Tableau dashboards, operational inefficiencies were identified and strategic recommendations were developed.